

Syntony: Practical Network Synchrony for Building Efficient Distributed Protocols

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Background

- Distributed protocols such as state machine replication form critical infrastructure for data center applications.
- Practical distributed protocols operate with **minimal synchrony assumptions about the underlying network to ensure correctness**, preventing them from fully leveraging available performance potential.

Motivation

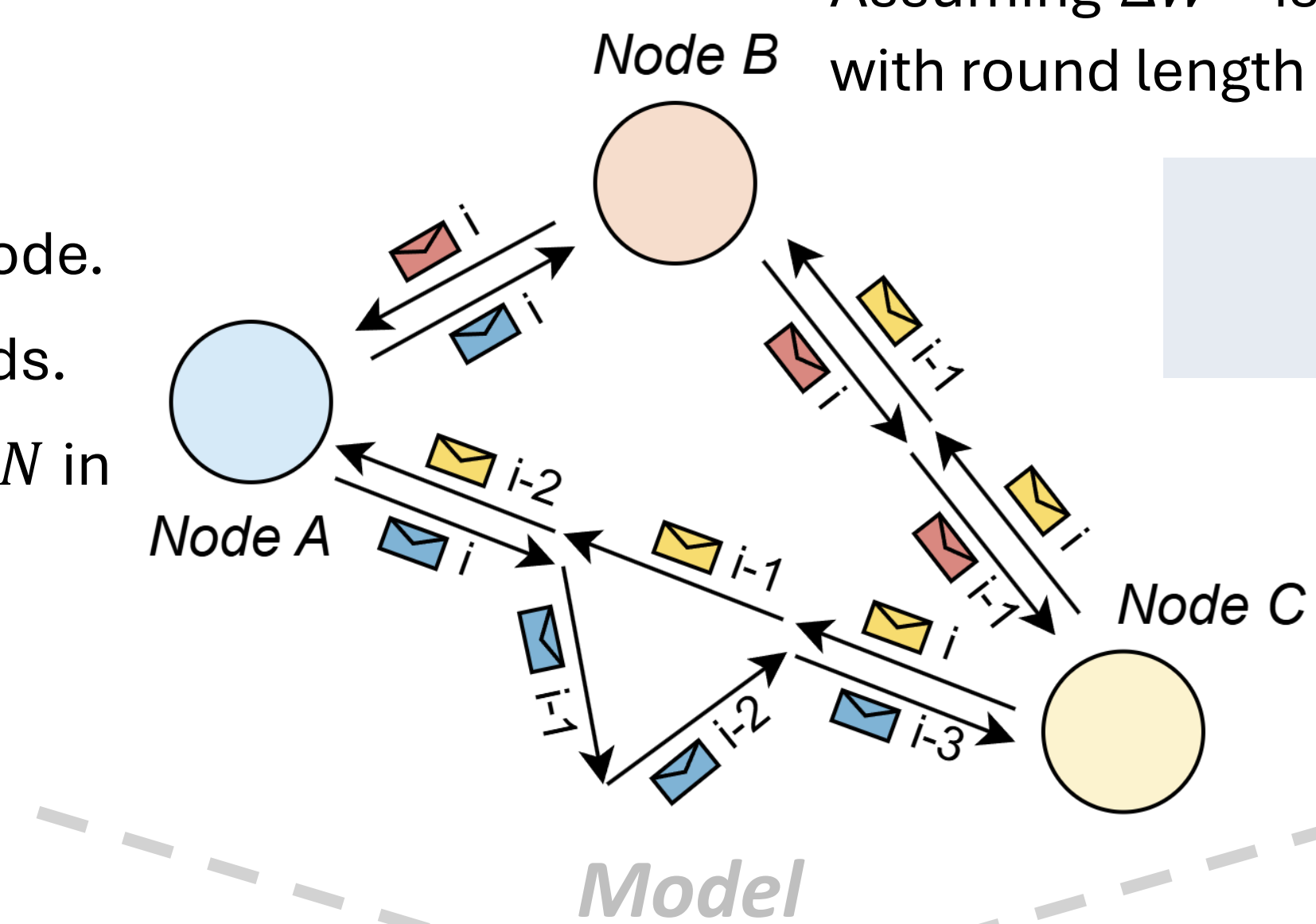
- Focusing on a stronger form of partial synchrony with **low-variance latencies**.
- Improving the performance** of distributed protocols under such strong synchrony.

The Syntony Network Model

Defining Syntony round to decouple propagation delays

In each Syntony round, for node N :

- N multicasts a message.
- N receives and processes one message from each peer node.
- N may receive and process messages from different rounds.
- Other nodes may receive and process messages sent by N in different rounds.



Methods

- Engineering **efficient lock-step rounds** whose lengths are mostly bounded by processing.
- Co-designing distributed protocols that perform **effective pipelined processing** on top of the engineered lock-step rounds.

Quantifying synchrony for performance

Determining the lower bound of round length that allows the system to complete round processing with a probability of $x\%$:

$$\Delta T_x \geq P_{x^{th}}(d_{prop}) - \mu(d_{prop}) + P_{x^{th}}(d_{proc})$$

Defining the x^{th} **Syntony coefficient** as:

$$S^x = \frac{\mu(d_{proc})}{P_{x^{th}}(d_{prop}) - \mu(d_{prop}) + P_{x^{th}}(d_{proc})}$$

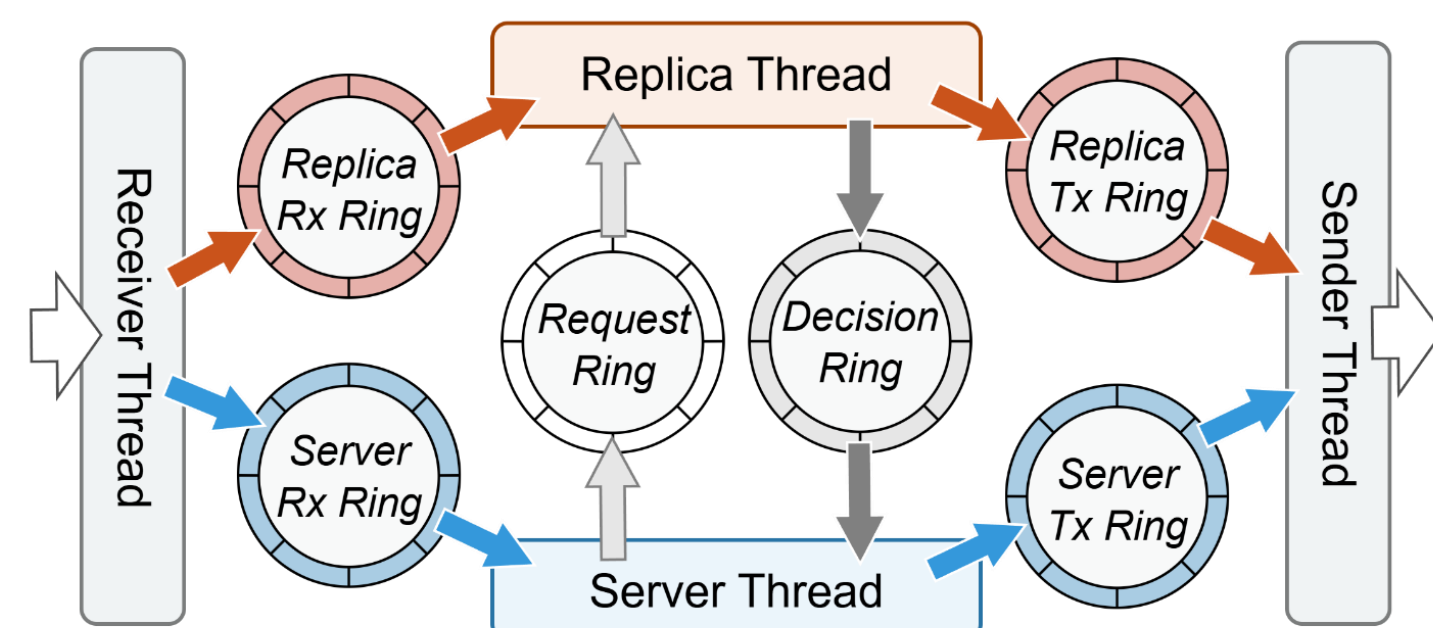
Assuming ΔW^x is the average workload that the system outputs in each round with round length ΔT_x , describing the upper bound of the system's throughput:

$$T_{put}^x = \frac{\Delta W^x}{\Delta T_x} \leq S^x \frac{\Delta W^x}{\mu(d_{proc})}$$

When $S^x \rightarrow 1$, throughput is only bounded by $\mu(d_{proc})$, representing a processing utility rate of 100%.

Engineering Host Networking

- Applying kernel-bypass network stack to reduce I/O processing latency.
- Isolating the critical path using a multithreaded architecture.
- Allowing early exit of a round either when having processing messages from all other replicas.



Co-Designing SMR Protocol

- Tolerating f crash failures with $2f + 1$ replicas.
- Assuming partial synchrony for correctness.
- Operating in **pulsing mode** during periods of strong synchrony and falling back to **responsive mode** to ensure liveness.
- Handling crashes with view changes.

Pulsing Mode:

Only Tx (broadcast) at start of rounds

Evaluation

Main Results

- Syntony improves throughput by over 100% compared to typical SMR protocols without compromising latency, when using the same DPDK-based network stack.
- The round length configuration reflects a trade-off between pipelined processing effectiveness and round efficiency and impacts system performance.
- Syntony shows better performance with stronger synchrony.

Work-in-Progress

- Implementing Syntony with RDMA UD multicasting and comparing it to state-of-the-art RDMA-based systems.
- Assessing the synchrony that can be achieved by the network infrastructure (as opposed to the end hosts) in different data center settings.

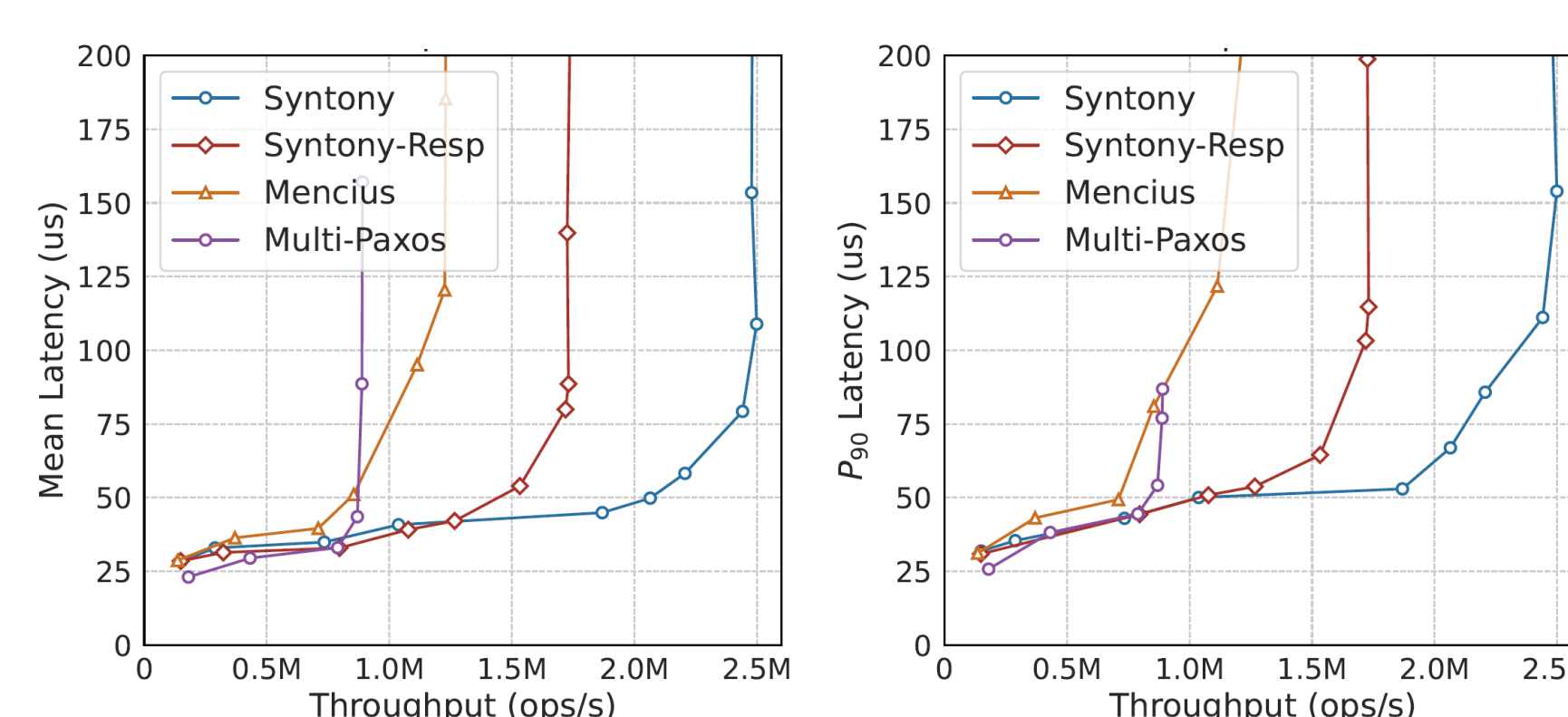


Figure 1. Latency-throughput graphs with 5 replicas.

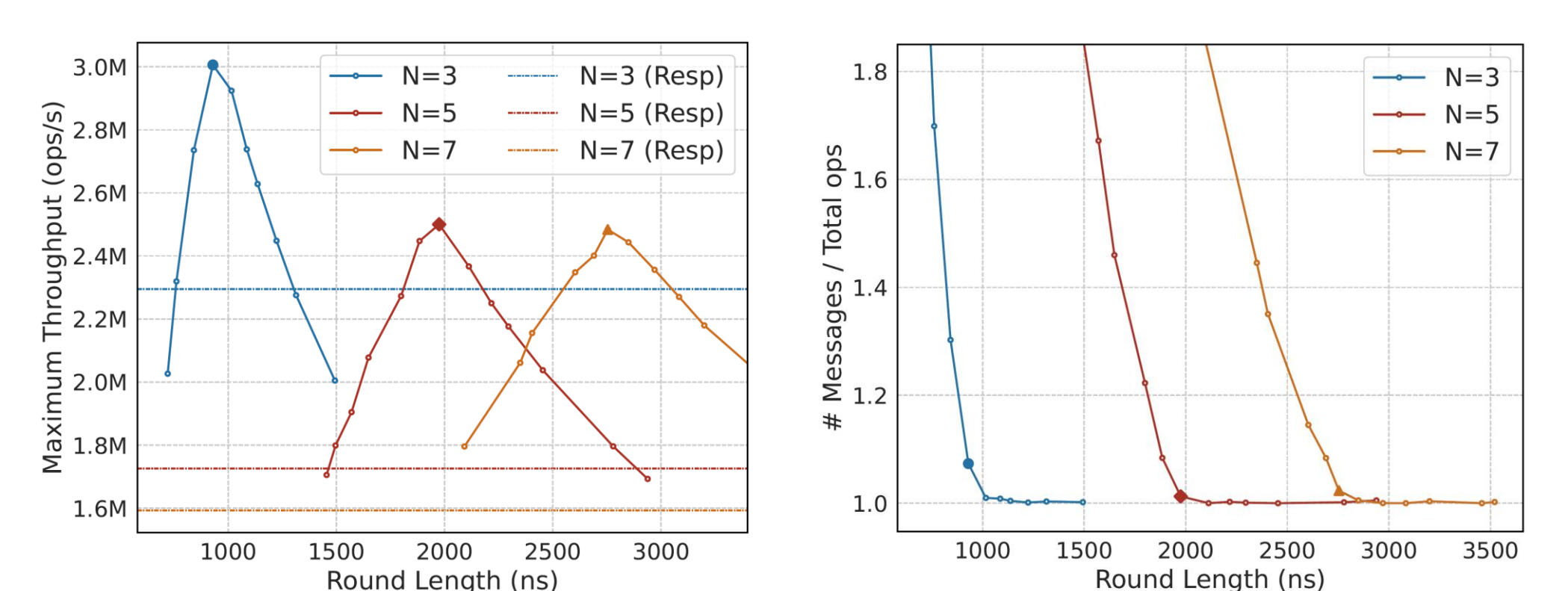


Figure 2. Impact of round length configuration on performance.

Table 1. Maximum throughput.

Mops/s	Number of Replicas		
	3	5	7
Multi-Paxos	1.52(-50%)	0.89(-64%)	0.69(-72%)
Mencius	1.64(-46%)	1.23(-51%)	1.17(-52%)
Syntony-Resp	2.30(-25%)	1.73(-31%)	1.59(-35%)
Syntony	3.01	2.50	2.44

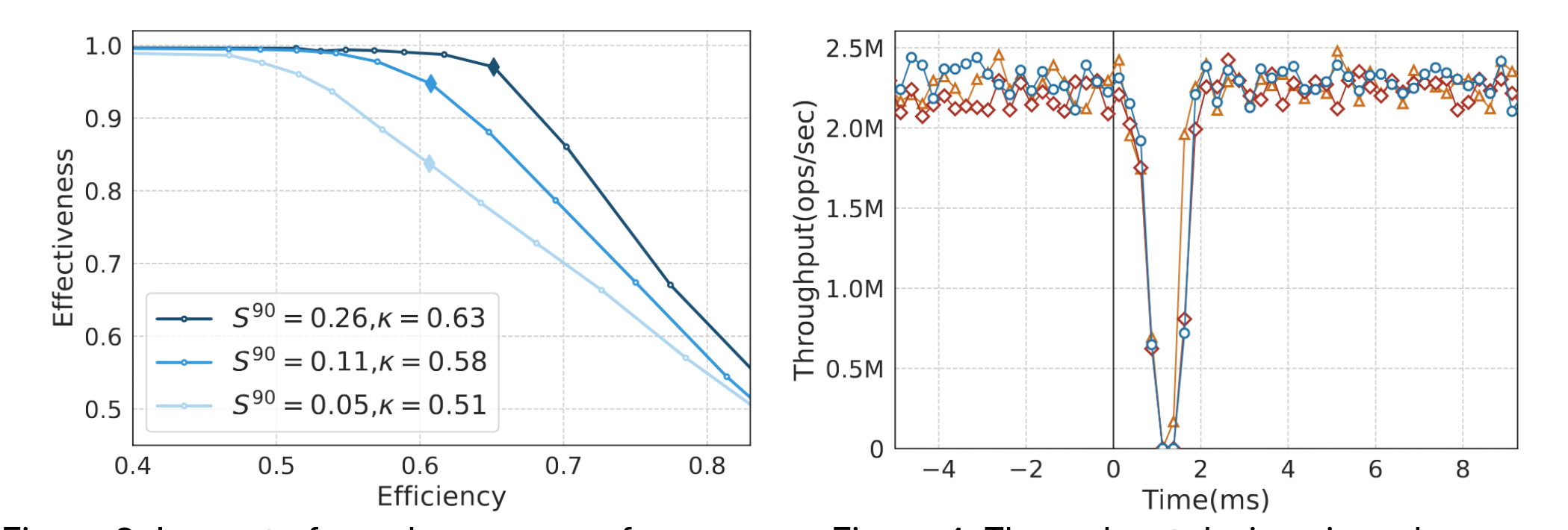


Figure 3. Impact of synchrony on performance.

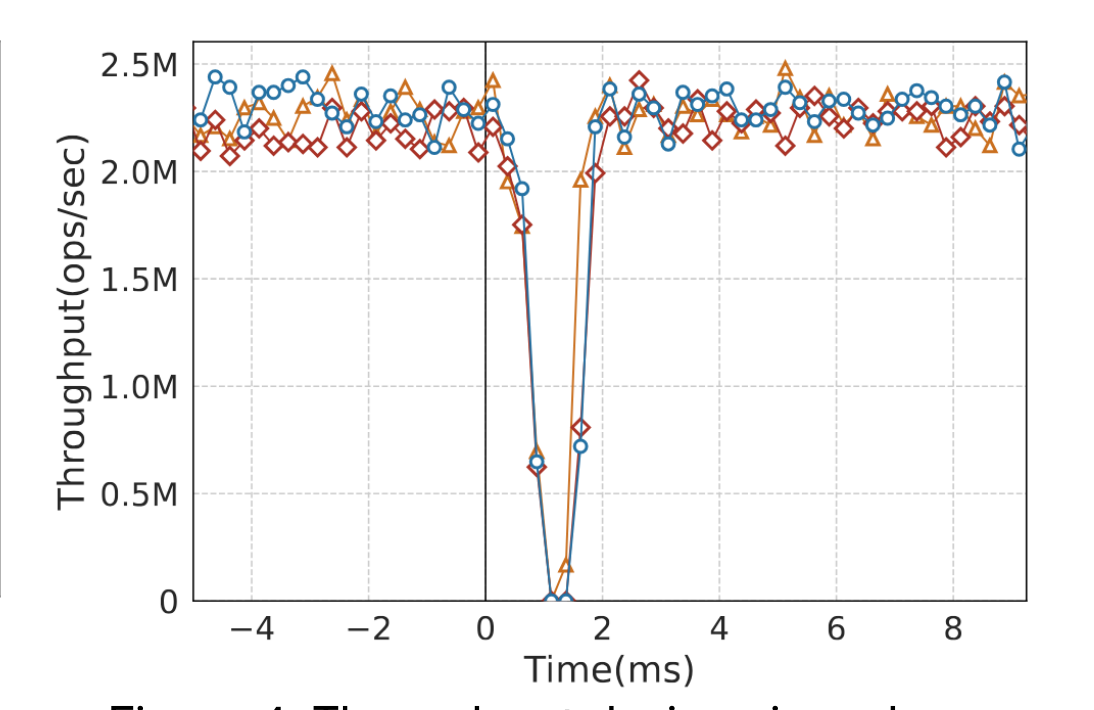


Figure 4. Throughput during view changes.

